

## **A MemComputing Approach to Orbital Mechanics**

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The MemComputing paradigm of computation employs time non-locality (memory) to solve a given problem by mapping its solutions into fixed points of a dynamical system. The problem is then solved not in the original state space of variables, but in an augmented phase space of relaxed original variables plus memory degrees of freedom. Although such an approach has been primarily used to solve combinatorial optimization problems, here we show that it can be employed in the field of celestial mechanics. As an important example we introduce a MemComputing machine that discovers all five Lagrange equilibrium points of any restricted three-body system in the solar system without prior knowledge of their location. The dynamical system follows equations of motion in the co-rotating frame with a dissipative term dynamically modified by a memory degree of freedom according to the gradient of the Jacobi effective potential. We finally discuss the advantages of this approach compared to existing methods.

## I. INTRODUCTION AND SCOPE

MemComputing is a non-von-Neumann paradigm in which a physical system exploits *time non-locality*—the ability of internal degrees of freedom to retain and use information about their past state—to solve a problem<sup>1</sup>. In a Digital MemComputing Machine (DMM) the memory variables act as dynamic Lagrange multipliers: the solutions of the target problem are mapped onto the stable fixed points of a continuous dynamical system, so that trajectories launched from generic initial conditions are guided to a solution along instanton paths<sup>2,3</sup>. The computation proceeds in an augmented phase space whose coordinates are the relaxed original variables *plus* the memory degrees of freedom. To date this paradigm has been applied mostly to combinatorial optimization (satisfiability, MaxSAT, prime factorization)<sup>4</sup>. Here we show that it transfers naturally to a continuous problem of celestial mechanics.

The restricted three-body problem (RTBP)—two massive primaries on a circular orbit and a massless test particle—is a cornerstone of nonlinear dynamics and of practical mission design<sup>5</sup>. Its five equilibria, the Lagrange points<sup>6</sup>, are normally located by analytic or numerical root finding that requires a separate bracket or initial guess for each point and prior knowledge of the solution structure. We construct instead a single MemComputing machine—an autonomous dynamical system in the co-rotating frame whose dissipation is dynamically controlled by one memory degree of freedom—that discovers all five Lagrange points of *any* restricted three-body system in the solar system from a generic grid of starting positions, without being given the coordinates of any solution. We verify it on all twenty-three Sun–planet and planet–moon pairs, spanning mass ratios from  $\mu \approx 2 \times 10^{-9}$  to  $\mu \approx 0.1$ , list every parameter needed to reproduce the results, and close by comparing the approach with existing methods.

## II. THE RESTRICTED THREE-BODY PROBLEM

We use the co-rotating frame in normalized units: the two primaries are separated by unit distance, the orbital angular velocity of the frame is  $\omega = 1$ , and the mass ratio is

$$\mu = \frac{M_2}{M_1 + M_2} \in (0, \tfrac{1}{2}], \quad (1)$$

placing the primary  $M_1$  at  $(-\mu, 0)$  and the secondary  $M_2$  at  $(1 - \mu, 0)$ . The Jacobi effective potential is

$$\Omega(x, y) = \frac{x^2 + y^2}{2} + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2}, \quad (2)$$

with  $r_1 = \sqrt{(x + \mu)^2 + y^2}$  and  $r_2 = \sqrt{(x - 1 + \mu)^2 + y^2}$ ; the first term is the centrifugal potential of the rotating frame and the other two are the gravitational wells of the primaries. Its gradient is

$$\partial_x \Omega = x - \frac{(1 - \mu)(x + \mu)}{r_1^3} - \frac{\mu(x - 1 + \mu)}{r_2^3}, \quad (3)$$

$$\partial_y \Omega = y - \frac{(1 - \mu)y}{r_1^3} - \frac{\mu y}{r_2^3}. \quad (4)$$

A Lagrange point is a relative equilibrium: in the rotating frame the test particle is stationary, so all velocities and accelerations vanish and the equilibrium condition reduces to the single vector clause

$$\boxed{\nabla \Omega(x, y) = \mathbf{0}} \quad (5)$$

which has exactly five solutions: the three collinear points  $L_1, L_2, L_3$  on the axis  $y = 0$  and the two triangular points  $L_{4,5} = (\frac{1}{2} - \mu, \pm \frac{\sqrt{3}}{2})^5$ .

A quantity central to the machine below is the transverse curvature of  $\Omega$ ,

$$\Omega_{yy} = 1 - \frac{1 - \mu}{r_1^3} + \frac{3(1 - \mu)y^2}{r_1^5} - \frac{\mu}{r_2^3} + \frac{3\mu y^2}{r_2^5}. \quad (6)$$

On the axis ( $y = 0$ ) it reduces to  $1 - (1 - \mu)/r_1^3 - \mu/r_2^3$ , which is *negative* at all three collinear points (for Earth–Moon,  $\mu = 0.0121$ :  $\Omega_{yy} = -4.15, -2.19, -0.011$  at  $L_1, L_2, L_3$ )

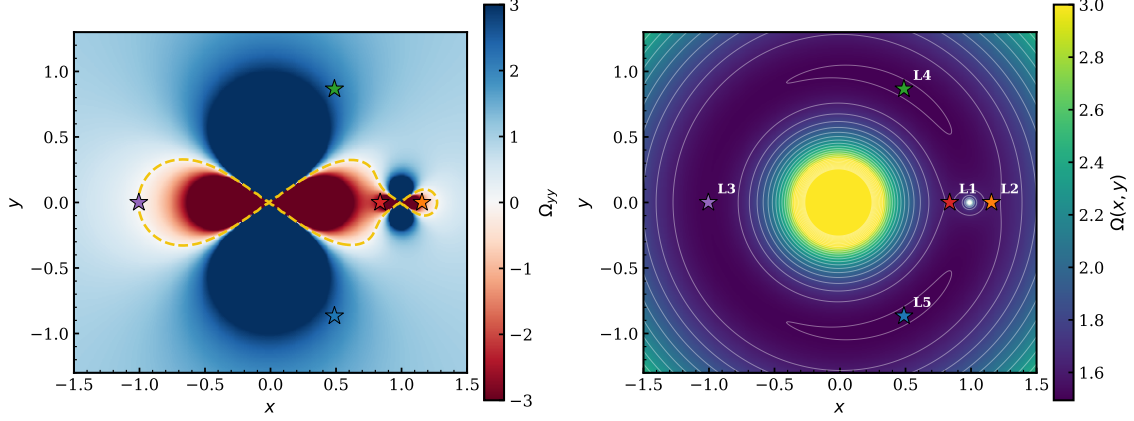


FIG. 1. **Left:** the transverse curvature  $\Omega_{yy}(x, y)$ , Eq. (6), for the Earth–Moon mass ratio (red–blue, clipped at  $\pm 3$ ); the dashed line is the contour  $\Omega_{yy} = 0$ . Where  $\Omega_{yy} < 0$  (red) the machine activates a correction current ( $\sigma = +1$ ); elsewhere it descends normally ( $\sigma = -1$ ). **Right:** the effective potential  $\Omega(x, y)$ , Eq. (2). Stars mark the five Lagrange points.

and *positive* ( $+2.25$ ) at  $L_{4,5}$ . Its sign therefore distinguishes the collinear saddle points from the triangular extrema using only information available at the current position (Fig. 1).

### III. THE MEMCOMPUTING MACHINE

The machine is an autonomous dynamical system in the co-rotating frame. To the ordinary rotating-frame equations of motion—the gradient force  $-\nabla\Omega$  and the Coriolis acceleration—we add a viscous term whose coefficient is *not constant* but is set by a memory degree of freedom  $m$ :

$$\ddot{x} = 2\dot{y} - \partial_x\Omega - \gamma_{\text{eff}}\dot{x}, \quad (7a)$$

$$\ddot{y} = -2\dot{x} + \sigma\partial_y\Omega - \gamma_{\text{eff}}\dot{y}, \quad (7b)$$

$$\dot{m} = \beta\|\nabla\Omega\|, \quad m(0) = 0, \quad m \leq m_{\text{cap}}, \quad (7c)$$

$$\gamma_{\text{eff}} = \gamma_0 + \kappa m. \quad (7d)$$

The augmented phase space is  $(x, y, \dot{x}, \dot{y}, m)$ : the relaxed original variables  $(x, y)$  and their velocities, plus the single memory variable  $m$ .

*a. Memory degree of freedom.* Equation (7c) integrates the magnitude of the clause violation,  $\|\nabla\Omega\|$ , so that

$$m(t) = \beta \int_0^t \|\nabla\Omega(\mathbf{r}(\tau))\| d\tau. \quad (8)$$

Because the integrand is non-negative,  $m$  is monotone non-decreasing: it ratchets up while the clause  $\nabla\Omega = \mathbf{0}$  is unsatisfied and stops growing only when  $\|\nabla\Omega\| \rightarrow 0$ , i.e. exactly at a Lagrange point. Through  $\gamma_{\text{eff}} = \gamma_0 + \kappa m$  this accumulated history controls the dissipation of the mechanical motion. The memory enters the dissipative channel deliberately: the rate of change of the kinetic energy  $T = \frac{1}{2}\|\dot{\mathbf{r}}\|^2$  along (7) is

$$\dot{T} = -\dot{\mathbf{r}} \cdot \nabla_{\sigma}\Omega - (\gamma_0 + \kappa m) \|\dot{\mathbf{r}}\|^2, \quad (9)$$

where  $\nabla_{\sigma}\Omega = (\partial_x\Omega, -\sigma\partial_y\Omega)$  and the Coriolis force has dropped out because it does no work. The only sign-definite, energy-removing term is the viscous one, and it is the memory  $m$  that sets its strength. With  $\gamma_0 = 0$  the memory is the *sole* source of dissipation: the flow is undamped at  $t = 0$  and acquires dissipation only as  $m$  accumulates from the history of the clause violation. The monotone functional (8) thus plays the role of a Lyapunov function, draining kinetic energy until the trajectory is quenched at a fixed point of (7), which by construction is a solution of the clause (5).

*b. Curvature-adaptive correction current.* At each step the machine evaluates the local curvature  $\Omega_{yy}$ , Eq. (6), and sets

$$\sigma = \begin{cases} +1 & \Omega_{yy} < 0 \quad (\text{correction current}), \\ -1 & \Omega_{yy} \geq 0 \quad (\text{ordinary descent}). \end{cases} \quad (10)$$

At a collinear point a small transverse displacement  $y > 0$  gives  $\partial_y\Omega < 0$  (because  $\Omega_{yy} < 0$ ), so the ordinary force  $-\partial_y\Omega > 0$  would push the particle *away* from the axis; setting  $\sigma = +1$  reverses it into a restoring force, turning the collinear saddle into an attractor of the damped flow. Where  $\Omega_{yy} > 0$  the ordinary descent already restores. The sign is read from the geometry at the current position; no information about which equilibria are saddles is

supplied. We use the full gradient norm in Eq. (7c) (a single scalar memory) so that both axes are damped equally; a per-axis memory would underdamp the transverse direction on the collinear axis, where  $|\partial_y \Omega| \approx 0$ , and the trajectory would drift past  $L_2, L_3$ .

#### IV. NUMERICAL IMPLEMENTATION AND PARAMETERS

Everything below is the procedure actually integrated in the accompanying code<sup>8</sup>; all parameters are collected in Table I so the results can be reproduced exactly.

*a. Integration.* Equations (7) are advanced with the explicit Euler method and a fixed step  $\Delta t$ . A singularity guard  $\varepsilon = 10^{-9}$  is added to  $r_1, r_2$ . A trajectory stops when  $\|\nabla \Omega\| < \theta$  (the convergence threshold), when it leaves the disk  $\|\mathbf{r}\| > 12$ , or after  $N_{\max}$  steps.

*b. Seed grid.* No solution coordinates are given to the integrator. The starting positions are generated from the *generic structure* of the RTBP only—the rotating-frame symmetry, the Hill scaling, and the equilateral configuration—using the Hill radius  $r_H = (\mu/3)^{1/3}$  and the secondary position  $x_s = 1 - \mu$ :

- (A) *on-axis seeds*  $(x_s \pm c r_H, 0)$  for  $c \in \{0.5, 0.7, 1.0, 1.4, 2.0, 3.0, 5.0, 8.0\}$ , which bracket the collinear  $L_1, L_2$  that lie within a Hill radius of the secondary on the axis  $y = 0$ ;
- (B)  *$L_3$  seeds* at  $(-1 - 5\mu/12, 0)$ ,  $(-0.96, 0)$ ,  $(-1.04, 0)$ ,  $(-1.10, 0)$ , opposite the primary;
- (C) *equilateral seeds*  $(\frac{1}{2} + \delta_x, \pm \frac{\sqrt{3}}{2} + \delta_y)$  for  $\delta_x \in \{-0.12, -0.04, 0.04, 0.12\}$ ,  $\delta_y \in \{-0.10, 0, 0.10\}$ , around the triangular configuration;
- (D) a coarse off-axis fill  $(x, y)$  with  $x \in \text{linspace}(-1.3, 1.3, 6)$  and  $y \in \{\pm 0.9, \pm 0.45\}$ , for the broad large- $\mu$  basins.

Seeds falling inside the exclusion disks around either primary—radius  $\varepsilon_1 = \min(0.03, 0.3 r_H)$  about  $M_1$  and  $\varepsilon_2 = \min(0.012, 0.3 r_H)$  about  $M_2$ —are discarded. This yields of order seventy starting positions per system. The seeds use no solution coordinates: the collinear points are transcendental roots whose  $\mu$ -dependent positions are not supplied, and even the equilateral seeds are placed at the fixed point  $(\frac{1}{2}, \pm \frac{\sqrt{3}}{2})$  rather than the exact  $(\frac{1}{2} - \mu, \pm \frac{\sqrt{3}}{2})$ ; the dynamics and the refinement below produce the exact positions.

*c. Refinement and labelling.* Each stopped trajectory is optionally refined by a few Newton iterations on  $\nabla\Omega = \mathbf{0}$  (using the analytic Hessian), accepted only if it converges to a genuine zero,  $\|\nabla\Omega\| < 10^{-9}$ , and otherwise falling back to the raw endpoint. A converged point is identified with the nearest of the five analytic Lagrange positions if it lies within 0.05 (refined) or 0.12 (raw); these analytic positions are used only for post-hoc identification and accuracy assessment, never by the integrator.

TABLE I. Parameters used to produce all results. Control parameters are the same for every system.

| Symbol           | Meaning                                  | Value                        |
|------------------|------------------------------------------|------------------------------|
| $\gamma_0$       | baseline damping                         | 0                            |
| $\kappa$         | memory–damping gain                      | 1.0                          |
| $\beta$          | memory growth rate                       | 0.5                          |
| $m_{\text{cap}}$ | memory cap                               | 10                           |
| $\Delta t$       | Euler time step                          | 0.01                         |
| $\theta$         | convergence threshold $\ \nabla\Omega\ $ | $10^{-4}$                    |
| $N_{\text{max}}$ | maximum steps                            | $2 \times 10^5$              |
| $\varepsilon$    | singularity guard on $r_1, r_2$          | $10^{-9}$                    |
| $\varepsilon_1$  | exclusion radius about $M_1$             | $\min(0.03, 0.3 r_H)$        |
| $\varepsilon_2$  | exclusion radius about $M_2$             | $\min(0.012, 0.3 r_H)$       |
| Newton xtol      | refinement tolerance                     | $10^{-12}$                   |
| accept           | refinement acceptance                    | $\ \nabla\Omega\  < 10^{-9}$ |

## V. RESULTS

Figure 2 shows representative trajectories of the machine for the Earth–Moon system: each is an instanton path from a generic seed to one of the five Lagrange points. Figure 3 tracks the internal dynamics along the converging trajectories: the memory  $m$  and the dissipation  $\gamma_{\text{eff}} = \kappa m$  it generates ratchet upward from zero and saturate, the kinetic energy  $T$  is driven to zero, and the clause violation  $\|\nabla\Omega\|$  decays below threshold. The memory is doing the work: with  $\gamma_0 = 0$  it is the only term that removes energy.

The central result is that the *same* machine, with the *same* control parameters of Table I, discovers all five Lagrange points of every two-body system in the solar system. Table II lists the twenty-three Sun–planet and planet–moon pairs, with mass ratios spanning eight

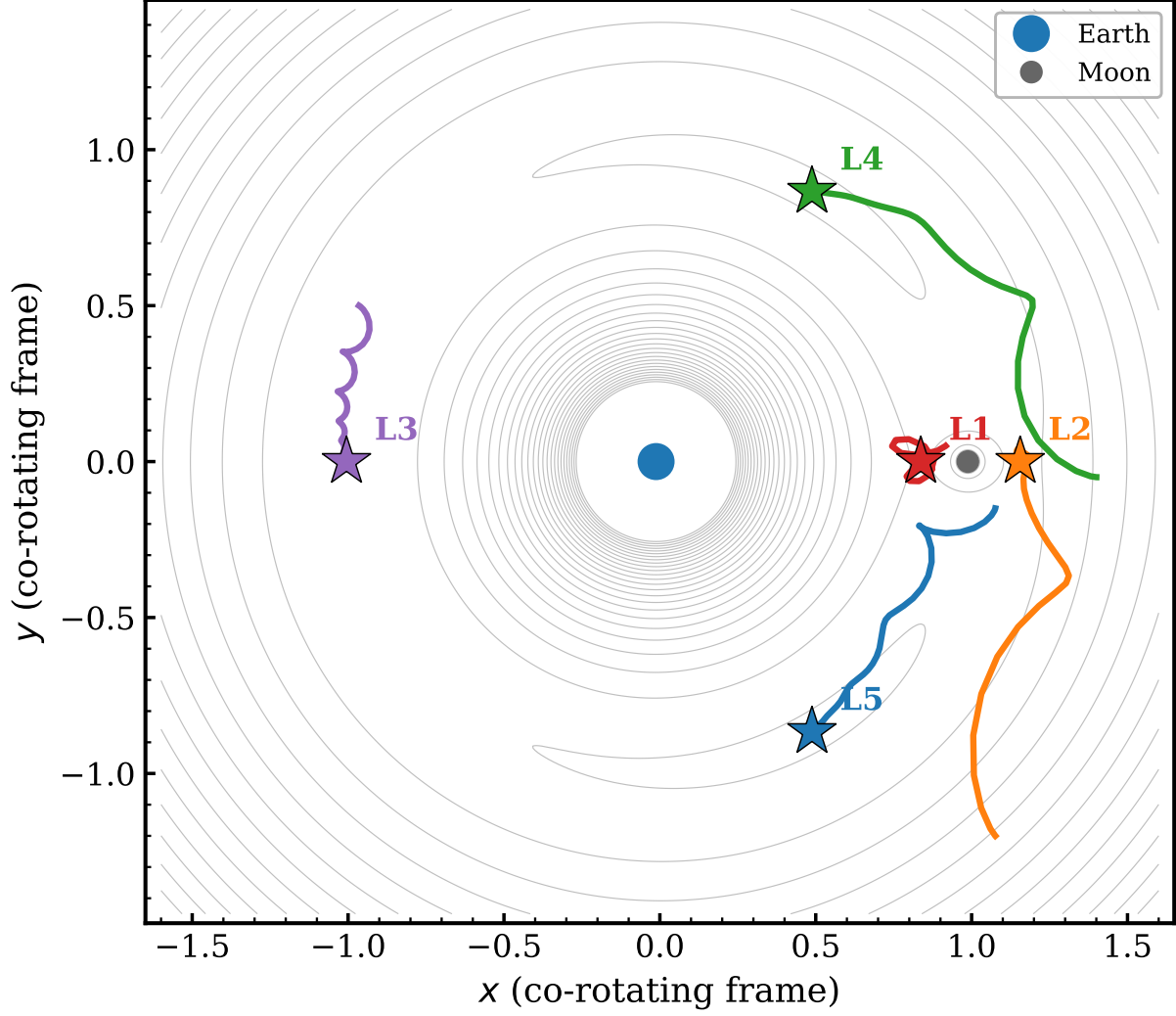


FIG. 2. Representative trajectories of the MemComputing machine for the Earth–Moon system ( $\mu = 0.0121$ ), one per Lagrange point. Grey curves: level sets of  $\Omega$ . Stars: the discovered equilibria. The integrator receives no solution coordinates.

orders of magnitude, from Mars–Deimos ( $\mu = 2.3 \times 10^{-9}$ ) to Pluto–Charon ( $\mu = 0.11$ ). Every system returns 5/5. After Newton refinement the located positions agree with the analytic Lagrange points to better than  $10^{-9}$ . We note that for  $\mu < \mu_{\text{Routh}} = 0.03852^7$  the triangular points  $L_4, L_5$  are linearly stable, satisfied by all systems except Pluto–Charon, where the machine still locates them as solutions of  $\nabla\Omega = \mathbf{0}$ .



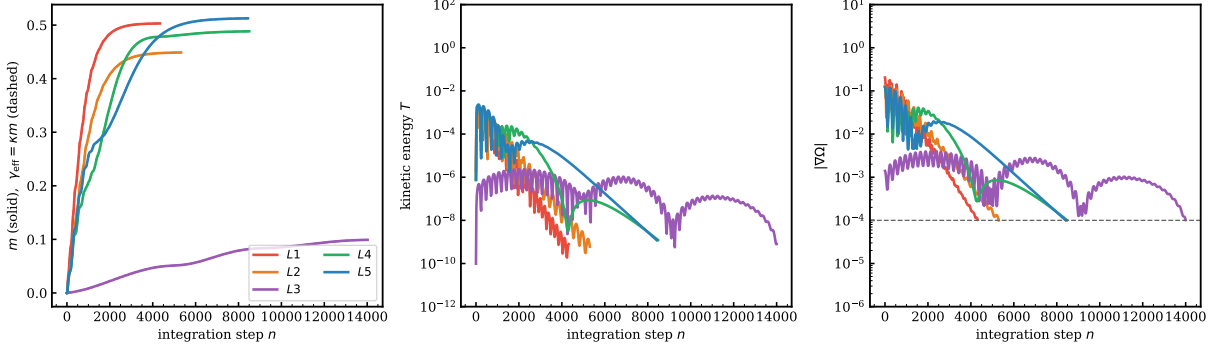


FIG. 3. Internal dynamics of the machine along the converging Earth–Moon trajectories. **Left:** memory  $m$  (solid) and the dissipation it generates  $\gamma_{\text{eff}} = \kappa m$  (dashed) ratchet up and saturate. **Centre:** kinetic energy  $T$  driven to zero (memory removes it). **Right:** clause violation  $\|\nabla\Omega\|$  decays below the threshold  $\theta = 10^{-4}$  (dashed). Colour identifies the discovered L-point.

## VI. ADVANTAGES OVER EXISTING METHODS

The Lagrange points can of course be obtained by classical means, and we are careful not to overstate the case. A Newton or bracketing iteration converges to a single point in a handful of evaluations once a good initial guess or bracket is supplied, and is far cheaper per point than integrating a trajectory. The advantages of the MemComputing machine lie elsewhere and are of a different kind.

First, it is *global and prior-knowledge-free in the solution*: a single autonomous flow, with one fixed parameter set, locates all five equilibria of any system from a generic structural grid, without a separate bracket or initial guess per point and without being told the collinear symmetry or any solution coordinate. Classical root finders must be targeted at each root individually and, for the collinear points, told that they lie on  $y = 0$ .

Second, it returns the solutions in the *natural MemComputing representation*—as the stable fixed points of a dynamical system in an augmented phase space—so the same machinery that has been applied to combinatorial optimization carries over unchanged to a continuous celestial-mechanics problem. The memory degree of freedom supplies a principled, monotone (Lyapunov-type) dissipative mechanism rather than an ad-hoc step-size rule.

Third, the construction is uniform across an enormous range of mass ratios: the *same*

TABLE II. All five Lagrange points are found for every solar-system two-body pair, using the single parameter set of Table I. “Stable” refers to the linear stability of  $L_4, L_5$  (Routh,  $\mu < 0.03852$ ).

| System           | $\mu$                | Found $L_4/L_5$ stable |     |
|------------------|----------------------|------------------------|-----|
| Mars–Deimos      | $2.3 \times 10^{-9}$ | 5/5                    | yes |
| Sun–Pluto        | $6.6 \times 10^{-9}$ | 5/5                    | yes |
| Mars–Phobos      | $1.7 \times 10^{-8}$ | 5/5                    | yes |
| Sun–Mercury      | $1.7 \times 10^{-7}$ | 5/5                    | yes |
| Saturn–Enceladus | $1.9 \times 10^{-7}$ | 5/5                    | yes |
| Sun–Mars         | $3.2 \times 10^{-7}$ | 5/5                    | yes |
| Sun–Venus        | $2.4 \times 10^{-6}$ | 5/5                    | yes |
| Sun–Earth        | $3.0 \times 10^{-6}$ | 5/5                    | yes |
| Saturn–Rhea      | $4.1 \times 10^{-6}$ | 5/5                    | yes |
| Jupiter–Europa   | $2.5 \times 10^{-5}$ | 5/5                    | yes |
| Uranus–Oberon    | $3.5 \times 10^{-5}$ | 5/5                    | yes |
| Uranus–Titania   | $4.1 \times 10^{-5}$ | 5/5                    | yes |
| Sun–Uranus       | $4.4 \times 10^{-5}$ | 5/5                    | yes |
| Jupiter–Io       | $4.7 \times 10^{-5}$ | 5/5                    | yes |
| Sun–Neptune      | $5.1 \times 10^{-5}$ | 5/5                    | yes |
| Jupiter–Callisto | $5.7 \times 10^{-5}$ | 5/5                    | yes |
| Jupiter–Ganymede | $7.8 \times 10^{-5}$ | 5/5                    | yes |
| Neptune–Triton   | $2.1 \times 10^{-4}$ | 5/5                    | yes |
| Saturn–Titan     | $2.4 \times 10^{-4}$ | 5/5                    | yes |
| Sun–Saturn       | $2.9 \times 10^{-4}$ | 5/5                    | yes |
| Sun–Jupiter      | $9.5 \times 10^{-4}$ | 5/5                    | yes |
| Earth–Moon       | $1.2 \times 10^{-2}$ | 5/5                    | yes |
| Pluto–Charon     | $1.1 \times 10^{-1}$ | 5/5                    | no  |

equations and parameters succeed from  $\mu \sim 10^{-9}$  to  $\mu \sim 10^{-1}$ , including the small- $\mu$  regime where the corotation ridge ( $\|\nabla\Omega\| \approx 0$  along  $r = 1$ ) makes the collinear points delicate—there the on-axis structural seeds, which exploit only the rotating-frame symmetry and the Hill scaling, give the flow the foothold it needs.

These properties make the approach a faithful and instructive transcription of the MemComputing paradigm into orbital mechanics, complementary to—rather than a replacement for—fast local root finders.

## VII. CONCLUSION

We have introduced a MemComputing machine for the restricted three-body problem: an autonomous dynamical system in the co-rotating frame whose dissipation is dynamically

modified by a single memory degree of freedom according to the gradient of the Jacobi effective potential, with a curvature-adaptive correction current that renders the collinear saddle points attractors. The machine discovers all five Lagrange points of every Sun–planet and planet–moon pair in the solar system—twenty-three systems spanning  $\mu \approx 2 \times 10^{-9}$  to 0.1—from a generic structural grid and with a single fixed parameter set, and the located positions are refined to machine precision. The work extends the MemComputing paradigm, hitherto applied chiefly to combinatorial optimization, into celestial mechanics, and provides a transparent, fully reproducible example in which the memory degree of freedom is the active ingredient of the computation. The code and an interactive implementation are available<sup>8</sup>.

## ACKNOWLEDGMENTS

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